

3.6> In an examination 30% of the candidates obtained marks below 40 and 10% of the candidates got above 75 marks. Assuming that the marks are normally distributed, find mean and standard deviation of the distribution.

sq Let X be a random variable that denotes the normal distribution such that $X \sim N(\mu, \sigma^2)$.

i) when $X = 40$.

$$P(X < 40) = 30\%$$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{40 - \mu}{\sigma}\right) = 0.3$$

$$\Rightarrow P(Z < -Z_1) = 0.3$$

$$\Rightarrow 0.5 - P(-Z_1 < Z < 0) = 0.3$$

$$\Rightarrow 0.5 - 0.3 = P(0 < Z < Z_1) \quad (\text{By symmetry})$$

$$\Rightarrow P(0 < Z < Z_1) = 0.2$$

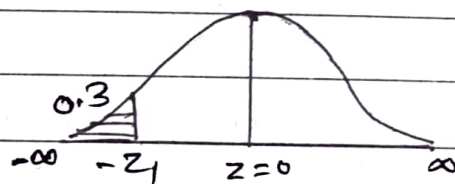
$$\Rightarrow P(0 < Z < 0.52) = 0.2$$

$$\therefore Z_1 = 0.52$$

$$\text{Since, } -Z_1 = \frac{40 - \mu}{\sigma} = 0.52$$

$$\Rightarrow -0.52 \sigma = 40 - \mu$$

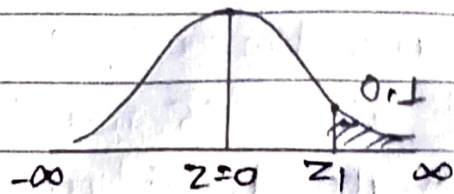
$$\therefore \mu - 0.52 \sigma = 40 \quad \text{--- (1)}$$



ii) When $X = 75$.

$$P(X > 75) = 10\% = 0.1$$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} > \frac{75 - \mu}{\sigma}\right) = 0.1$$



$$\Rightarrow P(Z > z_1) = 0.1$$

$$\Rightarrow 0.5 - P(0 < Z < z_1) = 0.1$$

$$\Rightarrow 0.5 - 0.1 = P(0 < Z < z_1)$$

$$\Rightarrow 0.4 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.4$$

$$\Rightarrow P(0 < Z < 1.28) = 0.4$$

$$\therefore z_1 = 1.28$$

Since,

$$z_1 = \frac{75 - \mu}{\sigma} = 1.28$$

$$\Rightarrow 75 - \mu = 1.28 \sigma$$

$$\therefore \mu + 1.28 \sigma = 75 \quad \text{--- (2)}$$

On solving (1) & (2), we get:-

$$\mu = 50.11 \approx 50 \quad \underline{\underline{Ans}}$$

$$\sigma = 19.44 \approx 19 \quad \underline{\underline{Ans}}$$

2022-Fall (P.V)

4. a) In a ~~quiz~~ class quiz, 8% of the students obtained marks below 25 and 90% of the students got below 85 marks. Assuming that the marks are normally distributed, find the mean and S.D. of the distribution.

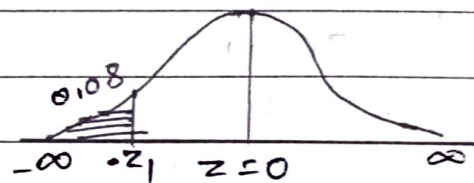
sol

Let X be a random variable that denotes the normal distribution such that $X \sim N(\mu, \sigma^2)$.

i) When $X = 25$

$$P(X < 25) = 8\%$$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{25 - \mu}{\sigma}\right) = 8\%$$



$$\Rightarrow P(Z < -z_1) = 0.08$$

$$\Rightarrow 0.5 - P(-z_1 < Z < 0) = 0.08$$

$$\Rightarrow 0.5 - 0.08 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.42$$

$$\Rightarrow P(0 < Z < 1.4) = 0.42 \quad (\text{From Normal table})$$

$$\therefore z_1 = 1.4$$

$$\text{Since, } -z_1 = \frac{25 - \mu}{\sigma}$$

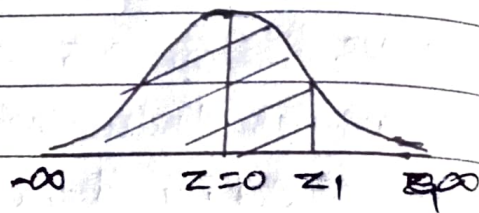
$$\Rightarrow -1.4 \sigma = 25 - \mu$$

$$\therefore \mu - 1.4 \sigma = 25 \quad \text{--- (1)}$$

ii) When $x = 85$

$$P(X \geq 85) = 90\%$$

$$\Rightarrow P\left(\frac{X-\mu}{\sigma} < \frac{85-\mu}{\sigma}\right) = 0.9$$



$$\Rightarrow P(Z < z_1) = 0.9$$

$$\Rightarrow 0.5 + P(0 < Z < z_1) = 0.9$$

$$\Rightarrow P(0 < Z < z_1) = 0.9 - 0.5$$

$$\Rightarrow P(0 < Z < z_1) = 0.4 \quad \text{or}$$

$$\Rightarrow P(0 < Z < 1.28) = 0.4 \quad (\text{From Normal table})$$

$$\therefore z_1 = 1.28$$

Since,

$$z_1 = \frac{85-\mu}{\sigma} = 1.28$$

$$\Rightarrow 85 - \mu = 1.28 \sigma$$

$$\therefore 1.28 \sigma + \mu = 85$$

$$\therefore \mu + 1.28 \sigma = 85 \quad \text{--- (2)}$$

On solving (1) & (2), we get:

$$\text{mean, } \mu = 56.34 \quad \text{Ans}$$

$$\text{s.d., } \sigma = 22.38 \quad \text{Ans}$$

2021-Spring

- 4.a) As a result of tests on 20,000 electric bulbs manufactured by a company, it was found that the life time of the bulb was normally distributed with an average of 2040 hours and the standard deviation of 60 hours. on the basis of the information,
- Estimate the number of bulbs that are expected to burn for less than 1960 hours.
 - Estimate the number of bulbs that are expected to burn in between 1860 hours to 2120 hours.
 - Find the minimum life time of 2,000 bulbs which have greater life time.

Given:-

No. of electric bulbs, $N = 20,000$.

Average, $\mu = 2040$ hrs

S.D., $\sigma = 60$ hrs

Let X be a random variable that denotes the binomial distribution such that $X \sim N(\mu, \sigma^2)$.

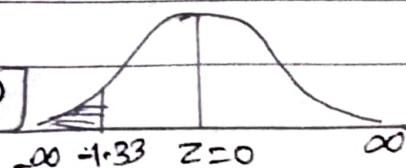
- i) The number of bulbs that are expected to burn for less than 1960 hours is given by

$$E(X < 1960) = N P(X < 1960)$$

$$= 20,000 \times P\left[\frac{X - \mu}{\sigma} < \frac{1960 - 2040}{60}\right]$$

$$= 20000 \times P(Z < -1.33)$$

$$= 20000 \times [0.5 - P(-1.33 < Z < 0)]$$



$$= 20000 \times [0.5 - p(0 < Z < 1.33)]$$

$$= 20000 \times (0.5 - 0.40824) \text{ (From Normal Table)}$$

$$= 1835.2$$

$$\approx 1835 \quad \underline{\text{Ans}}$$

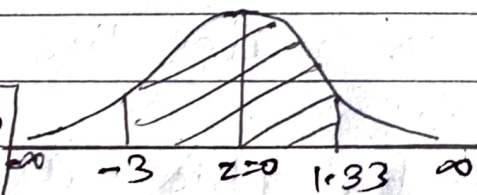
ii) The number of bulbs that are expected to burn in between 1860 hours to 2120 hours is given by

$$E(1860 < X < 2120)$$

$$= NP(1860 < X < 2120)$$

$$= NP \left[\frac{1860 - 2040}{60} < \frac{X - \mu}{\sigma} < \frac{2120 - 2040}{60} \right]$$

$$= NP[-3 < Z < 1.33]$$



$$= N[P(-3 < Z < 0) + P(0 < Z < 1.33)]$$

$$= N[P(0 < Z < 3) + P(0 < Z < 1.33)]$$

$$= 20000 (0.49865 + 0.40824)$$

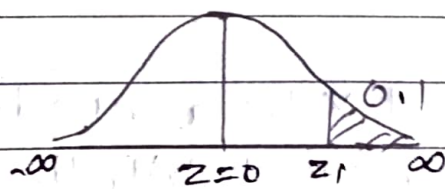
$$= 18137.8$$

$$\approx 18138 \quad \underline{\text{Ans}}$$

iii) The minimum lifetime of 2000 bulbs which have greater life time is given by

~~Let~~
Let X_1 be the minimum life time of bulbs.

~~$P(X < X_1) = \frac{2000}{20000}$~~
 $\therefore P(X < X_1) = \frac{2000}{20000}$



$$\Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{X_1 - \mu}{\sigma}\right) = 0.1$$

$$\Rightarrow P(Z < z_1) = 0.1$$

$$\Rightarrow 0.5 - P(0 < Z < z_1) = 0.1$$

$$\Rightarrow 0.5 - 0.1 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.4$$

$$\Rightarrow P(0 < z_1 < 1.28) = 0.4 \quad (\text{From Normal table})$$

$$\therefore z_1 = 1.28$$

Since, $z_1 = \frac{X_1 - \mu}{\sigma}$

$$\Rightarrow 1.28 = \frac{X_1 - 2040}{60}$$

$$\Rightarrow 1.28 \times 60 + 2040 = X_1$$

$$\therefore X_1 = 2116.8$$

Hence the minimum life time of 2000 bulbs which have greater life time is 2116.8 hr.

2021-Fall

4.a) In an experiment, 15% of the candidates got first class (60% mark or above), while 40% failed (scoring below 40% marks). Assuming the marks to be normally distributed, estimate the mean and standard deviation.

sol

Let X be a random variable that denotes the normal distribution such that $X \sim N(\mu, \sigma^2)$.

i) when $X = 60$

$$\text{i.e. } P(X > 60) = 15\% = 0.15$$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} > \frac{60 - \mu}{\sigma}\right) = 0.15$$

$$\Rightarrow P(Z > z_1) = 0.15$$

$$\Rightarrow 0.5 - P(0 < Z < z_1) = 0.15$$

$$\Rightarrow 0.5 - 0.15 = P(0 < Z < z_1)$$

$$\Rightarrow 0.35 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.35$$

$$\Rightarrow P(0 < Z < 1.04) = 0.35 \quad (\text{From Normal table})$$

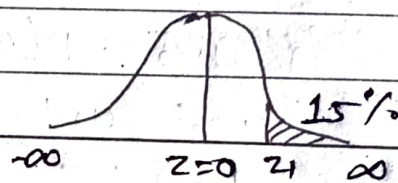
$$\therefore z_1 = 1.04$$

since,

$$z_1 = \frac{60 - \mu}{\sigma} = 1.04$$

$$\Rightarrow 60 - \mu = 1.04\sigma$$

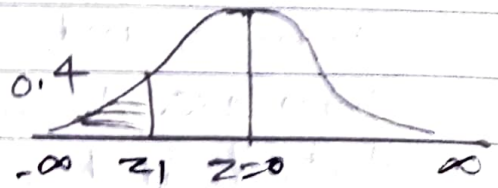
$$\therefore \mu + 1.04\sigma = 60 \quad \text{--- (1)}$$



ii) When $X = 60$

$$\text{i.e. } P(X < 40) = 40\% = 0.4$$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{40 - \mu}{\sigma}\right) = 0.4$$



$$\Rightarrow P(Z < -z_1) = 0.4$$

$$\Rightarrow 0.5 - P(-z_1 < Z < 0) = 0.4$$

$$\Rightarrow 0.5 - 0.4 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.1$$

$$\Rightarrow P(0 < Z < 0.25) = 0.1$$

$$\Rightarrow P(0 < Z < 0.25) = 0.1$$

$$\therefore z_1 = 0.25$$

$$\text{since, } -z_1 = \frac{40 - \mu}{\sigma} = -0.25$$

$$\Rightarrow 40 - \mu = -0.25\sigma$$

$$\therefore \mu - 0.25\sigma = 40 \quad \text{--- (2)}$$

On solving (1) & (2), we get:

$$\mu = 43.876$$

Ans

$$\sigma = 15.504$$

Ans

2020-Fall

4.6> In an examination, 25% of the students got marks more than 80 and 30% of the students got marks less than 45. Assuming the marks to be normally distributed, find mean and standard deviation of the marks of the students.

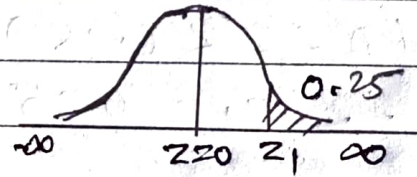
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Let X be a random variable that denotes the normal distribution such that $X \sim N(\mu, \sigma^2)$.

i> When $X = 80$

i.e. $P(X > 80) = 25\%$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} > \frac{80 - \mu}{\sigma}\right) = 25\% = 0.25$$



$$\Rightarrow P(Z > z_1) = 0.25$$

$$\Rightarrow 0.5 - P(0 < Z < z_1) = 0.25$$

$$\Rightarrow 0.5 - 0.25 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.25$$

$$\Rightarrow P(0 < Z < 0.675) = 0.25$$

$$\therefore z_1 = 0.675$$

Since,

$$z_1 = \frac{80 - \mu}{\sigma} = 0.675$$

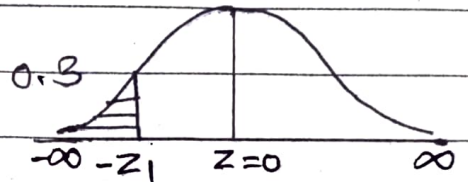
$$\Rightarrow 0.675 \sigma = 80 - \mu$$

$$\therefore \mu + 0.675 \sigma = 80 \quad \text{--- (1)}$$

ii) When $X = 45$.

i.e. $P(X < 45) = 30\%$.

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{45 - \mu}{\sigma}\right) = 0.3$$



$$\Rightarrow P(Z < -z_1) = 0.3$$

$$\Rightarrow 0.5 - P(-z_1 < Z < 0) = 0.3$$

$$\Rightarrow 0.5 - 0.3 = P(0 < Z < z_1)$$

$$\Rightarrow 0.2 = P(0 < Z < z_1)$$

$$\Rightarrow P(0 < Z < z_1) = 0.2$$

$$\Rightarrow P(0 < Z < 0.52) = 0.2$$

$$\therefore z_1 = 0.52$$

Since,

$$-z_1 = \frac{45 - \mu}{\sigma} = -0.52$$

$$\Rightarrow 45 - \mu = -0.52 \sigma$$

$$\therefore \mu - 0.52 \sigma = 45 \quad \text{--- (2)}$$

On solving (1) & (2), we get;

mean, $\mu = 60.23$ ~~not~~

S.D, $\sigma = 29.28$ ~~not~~